

Bernice Mercy Sharon Malaiarasu

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Profile

Generative AI engineer with 2+ years building production LLM systems at Thoughtworks, recipient of the Bayer GenAI Award for Best Technical Implementation. Built PRINCE, a multi-agent platform embedded in Bayer's preclinical research workflow across RAG, Text-to-SQL, and NER at scale. Now pursuing an MS in AI at Northeastern's Khoury College and seeking a 2026 Agentic AI co-op.

Education

Northeastern University – Khoury College of Computer Sciences	Boston, MA
MS in Artificial Intelligence, Specialization in Machine Learning	Jan. 2026 – May 2028
Loyola ICAM College of Engineering and Technology	Chennai, India
BE, Electronics and Communications Engineering	Aug. 2018 – May 2022

Awards & Talks

Bayer GenAI Award Best Technical Implementation for multi-agent GenAI system at Bayer	2025
Frontiers in AI Contributions to PRINCE acknowledged in peer-reviewed AI journal	2025
Speaker, XConf Multi-agent architecture of PRINCE and LLM evaluation strategies, Thoughtworks Bangalore	Aug 2025
GenAI Workshop Facilitator Designed and delivered workshop on LLMs and RAG to 50+ developers	Mar 2025
Speaker, Geeknight Talk on improving Text-to-SQL accuracy using dynamic few-shot prompting, 80+ engineers	Dec 2024

Experience

AIMES Lab, Northeastern University	Boston, MA
Research Assistant	Jul 2026 – Present
Husky AI – Platform enabling students to practice and refine prompt engineering skills through interactive exercises and guided feedback. AI Journalism Platform – Research project applying data analytics, AI video generation, and automated report generation to make news more accessible and reduce journalists' production workload. Lab WordPress Site – Maintain and extend the lab's WordPress website, implementing new features and resolving issues.	
Thoughtworks	Chennai, India
Sr. Consultant – Data Engineer	Aug 2022 – Dec 2025
PRINCE Chatbot – Generative AI	Jan 2024 – Dec 2025
- Led development of PRINCE (LangChain, LangGraph) — a ReAct-based multi-agent system with tool calling, deployed in production at Bayer across 18,000+ preclinical safety studies, cutting query response time by 30% and automating report generation. - Built the RAG backbone: 100 GB+ of biomedical data indexed in OpenSearch (vector store) using text embeddings, hybrid semantic + keyword search, 5x query expansion, and cross-encoder re-ranking; every response citation-linked to source. - Built Text-to-SQL on Claude 3.5 Sonnet using few-shot prompt engineering and structured output validation; pushed SQL accuracy above 90% against AWS Athena, cutting ad-hoc data requests from days to minutes. - Built a NER pipeline achieving 98%+ accuracy across 40+ entity fields, cutting data correction from days to 15 minutes and surfacing entry errors in 30%+ of records missed in prior manual review. - Built RAGAS + DeepEval as a CI/CD gate for MLOps; auto-ran expert-curated test sets on every code or prompt change across 5 metrics, with Langfuse observability and score drops triggering root-cause analysis before reaching production.	
PRINCE Data Platform – Data Engineering	Aug 2022 – Dec 2023
- Processed 20,000+ preclinical study PDFs — extracting, chunking, and generating text embeddings into OpenSearch as the vector store powering PRINCE's RAG retrieval. - Migrated 56 database tables from Kubernetes to AWS; built Terraform-managed pipelines syncing data into Elasticsearch as the query layer for the preclinical research API. - Built and maintained daily Spark (Scala) + AWS Glue ETL pipelines across 70+ source and 20+ sink tables, ingesting study reports, LIMS data, and lab metadata.	

Projects

E-Commerce Delivery Delay Prediction Northeastern	Apr 2026
XGBoost delivery delay predictor on 100K+ OLIST orders: 0.922 ROC-AUC and 6.8% validation delay rate. Used SHAP to identify which features were actually driving delays.	
Self-Harm Prevention Classifier Northeastern	June 2026
Fine-tuning LLaMA 3 on DeepSuiMind, a published benchmark of 1,308 implicit suicidal ideation cases with a standardized rubric; building a classification pipeline for early-intervention systems.	
Personal Assistance System for Visually Impaired Loyola ICAM	Nov 2021 – May 2022
Raspberry Pi assistive device combining Haar Cascade face recognition (96.24% accuracy), SSD object detection at 1s latency trained on 500+ images across 75 people, and ultrasonic obstacle sensing to 4m with real-time Bluetooth audio feedback.	

Technical Skills

AI/ML: LLMs, RAG Pipelines, Agentic AI, Multi-agent Systems, LangChain, LangGraph, Langfuse, RAGAS, DeepEval
Cloud: AWS (Glue, Athena, OpenSearch, DynamoDB, S3, Step Functions)
Languages & Frameworks: Python, Scala, SQL, PyTorch, TensorFlow, HuggingFace, OpenCV, Databricks/Spark, Terraform, Docker, Git